A NONTRIVIAL UNIVERSAL HOMEOMORPHISM OF SPECTRA YIELDS EITHER SQUARE AND CUBE RELATIONS OR A PRIME AND A P-TH POWER RELATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-pth-power-and-multiple is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 11656-11663 and proof span 11665-11710 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. Tag self-check per LANE_BRIEF 10.5 (5): the label lemma-pth-power-and-multiple resolves in the project's own tag list to canonical tag 0CNC (line 15887 of the index, full label morphisms-lemma-pth-power-and-multiple). reference/public-items.jsonl holds no row whose primary source_file is morphisms.tex, so the intersection with the same-chapter public tag ids is empty and the candidate passes.

Editorial note. The gate was run as: python operations/agent-authoring/tools/precard_gate.py –source morphisms.tex –label lemma-pth-power-and-multiple –start 11656 –end 11663 –proof-start 11665 –proof-end 11710 –local-id 34092, giving RESULT=OK with all four span guards passing, no public interval row for this chapter and occupancy FREE for morphisms.tex 11656 11710. Spans were recomputed live from the label in the pinned chapter file and the located text matched the pool byte counts. The interval lies outside the two windows of this chapter handed to lane-s1 (8000-10000 and 12000-14000).

Editorial note. 9 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A ring extension inducing a universal homeomorphism on spectra is integral, cited from the same chapter. Cited at proof lines 11671-11672. Gives integrality, which drives the reduction to a single generator and finiteness.
- Passing to a subalgebra generated by one element when testing the conclusion, and the closedness and nonemptiness of the support of a nonzero finite module, cited from the same chapter and the algebra chapter. Cited at proof lines 11673-11679. Reduces to a finite extension and produces a nonempty closed support whose minimal prime can be localized at.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- The conclusion is preserved by localizing the extension at a prime, cited from the same chapter. Cited at proof lines 11680-11683. Allows the reduction to a local base with the support of the quotient equal to the maximal ideal.
- Two consecutive powers of an element lying in the base imply the square and cube alternative, cited from the same chapter. Cited at proof lines 11700-11701. Delivers the first alternative of the conclusion when the prime of the extension is not contained in the base.
- A universally injective ring map with a nontrivial purely inseparable residue field extension has prime characteristic p with the extension purely inseparable, cited from the same chapter. Cited at proof lines 11704-11706. Produces the prime p and the element b of the second alternative in the remaining case.

Source reference. Original lemma 02JJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3338–3338. Same-chapter reference used in the retained author text at source line(s) 11029, 11628, 11676. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 00L2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 772–772. Same-chapter reference used in the retained author text at source line(s) 705, 759, 11631, 11640, 11679, 11688. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0585: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6768–6768. Same-chapter reference used in the retained author text at source line(s) 11631, 11679. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01WM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2487–2487. Same-chapter reference used in the retained author text at source line(s) 10976, 11172, 11375, 11380, 11512, 11552, 11624, 11672, 11769, 14152, 14165, 14544. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0CNA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15885–15885. Same-chapter reference used in the retained author text at source line(s) 11648, 11696. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0CN9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15884–15884. Same-chapter reference used in the retained author text at source line(s) 11635, 11683, 14852. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0CNB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15886–15886. Same-chapter reference used in the retained author text at source line(s) 11667, 11732, 11808. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0CN7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15882–15882. Same-chapter reference used in the retained author text at source line(s) 11625, 11673, 11931. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01S4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2325–2325. Same-chapter reference used in the retained author text at source line(s) 1676, 11525, 11564, 11702, 11781, 11892, 14730. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $A \subset B$ be a ring extension such that $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is a universal homeomorphism. If $A \neq B$, then either there exists a $b \in B$, $b \notin A$ with $b^2 \in A$ and $b^3 \in A$ or there exists a prime number p and a $b \in B$, $b \notin A$ with $pb \in A$ and $b^p \in A$.*

Proof. The argument is almost exactly the same as in the proof of Lemma 0CNB but we write everything out to make sure it works.

Recall that $A \subset B$ is integral (Lemma 01WM). By Lemma 0CN7 we may assume that B is generated by a single element over A. Hence B is finite over A (Algebra, Lemma 02JJ). Hence the support of B/A as an A-module is closed and not empty (Algebra, Lemmas 00L2 and 0585). Let $\mathfrak{p} \subset A$ be a minimal prime of the support. After replacing $A \subset B$ by $A_\mathfrak{p} \subset B_\mathfrak{p}$ (permissible by Lemma 0CN9) we may assume that $(A, \mathfrak{m})$ is a local ring, that B is finite over A, and that B/A has support $\{\mathfrak{m}\}$ as an A-module. Since B/A is a finite module, we see that $I = \mathrm{Ann}_A(B/A)$ satisfies $\mathfrak{m} = \sqrt{I}$ (Algebra, Lemma 00L2). Let $\mathfrak{m}' \subset B$ be the unique prime ideal lying over $\mathfrak{m}$. Because $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is a homeomorphism, we find that $\mathfrak{m}' = \sqrt{IB}$. For $f \in \mathfrak{m}'$ pick $n \geq 1$ such that $f^n \in IB$. Then also $f^{n+1} \in IB$. Since $IB \subset A$ by our choice of I we conclude that $f^n, f^{n+1} \in A$. Using Lemma 0CNA we conclude our lemma is true if $\mathfrak{m}' \not\subset A$. If $\mathfrak{m}' \subset A$, then $\mathfrak{m}' = \mathfrak{m}$. Since $A \neq B$ we conclude the map $\kappa = A/\mathfrak{m} \to B/\mathfrak{m}' = \kappa'$ of residue fields cannot be an isomorphism. By Lemma 01S4 we conclude that the characteristic of κ is a prime number p and that the extension κ'/κ is purely inseparable. Pick $b \in B$ whose image in κ' is an element not contained in κ but whose pth power is in κ. Then $b \notin A$, $b^p \in A$, and $pb \in A$ (because $pb \in \mathfrak{m}' = \mathfrak{m} \subset A$) as desired. □